MSD Song Proposal

Proposal

Eileen Ip

INTRODUCTION

Stakeholders:

The Untitled Group is Australia's leading concert organiser specialising in large-scale music festivals featuring top-tier artists. This report is primarily designed for the organisation's executives, event planners, and marketing strategists who are responsible for making data-driven decisions about artist selection and event planning.

Purpose:

The objective of this report is to enhance audience engagement and maximise concert success by identifying and recommending the most popular artists and their top-performing songs. Specifically, the report aims to visualise a diverse range of variables, analysing the key trends and offer actionable provide actionable insights. By conducting this, the report will optimise the artist selection process, artist's songs recommendation and identify current song format trends.

Benefits:

Leveraging data-driven insights offers several strategic advantages for the Untitled Group:

- Selecting the most popular artists ensures high ticket demand, directly impacting revenue potential.
- Featuring top artists increases the likelihood of audience enjoyment, engagement and encourages repeat attendance.
- Higher audience engagement can attract more sponsors and create valuable brand partnerships.
- Gain a competitive edge by leveraging data to identify trends and audience preferences.
- The actionable insights found from this project can refine future event strategies, improving artist selections and overall concert experiences.

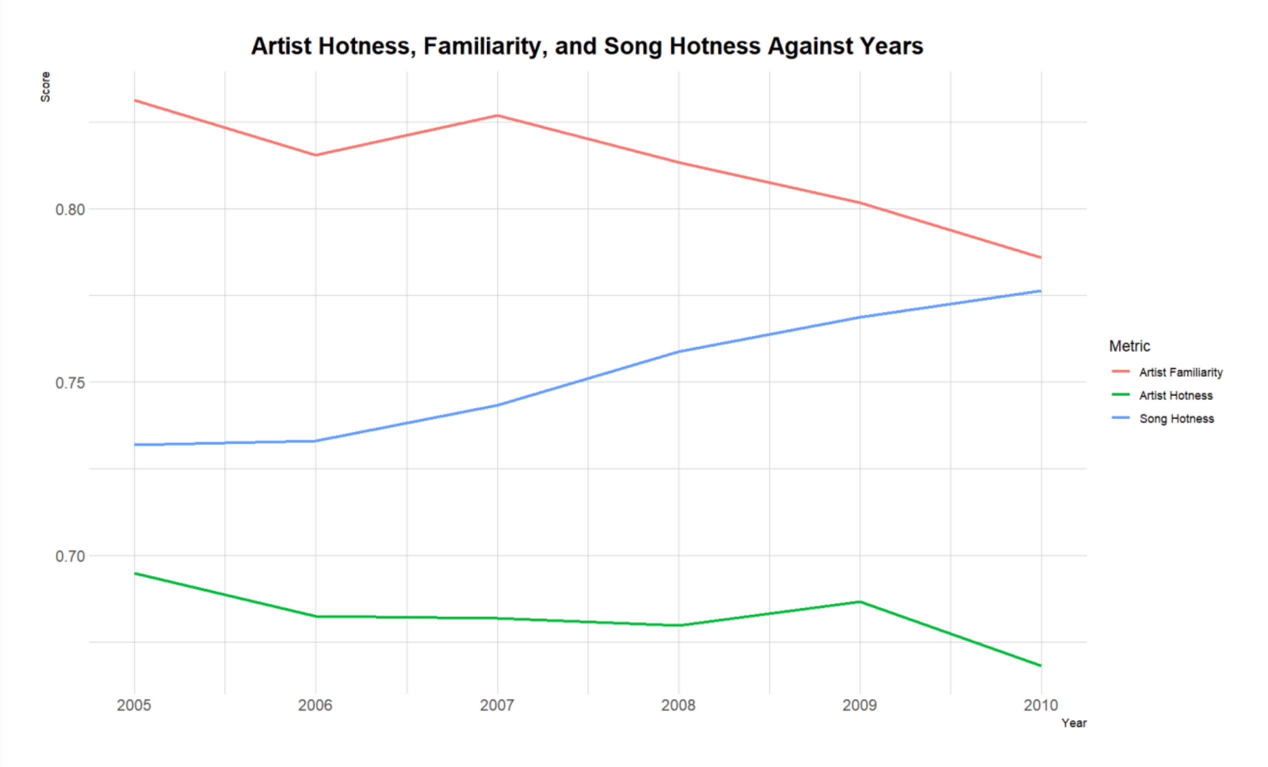
Key Variables:

The dataset, a subset of the million song dataset, encompasses one million song records. The following variables will be explored:

- Artist Name: Identify artists by name.
- Artist Familiarity: Evaluate artist recognition at that given time.
- Artist Hotness: Evaluate artist popularity at that given time.
- Song Hotness: Determine the popularity of individual songs.
- Duration: Ensure the durations of the songs are under 4 minutes for fair performance time.
- End of Fade In, Start of Fade Out, Key, Loudness, Tempo, and Time Signature: Analyse songs' formats and identify their common patterns over the years.
- Mode: Ensure the songs are in major chords to sound
- Year: To track artist performance consistency over time.

PLOTS

Chart 1: Line Plot

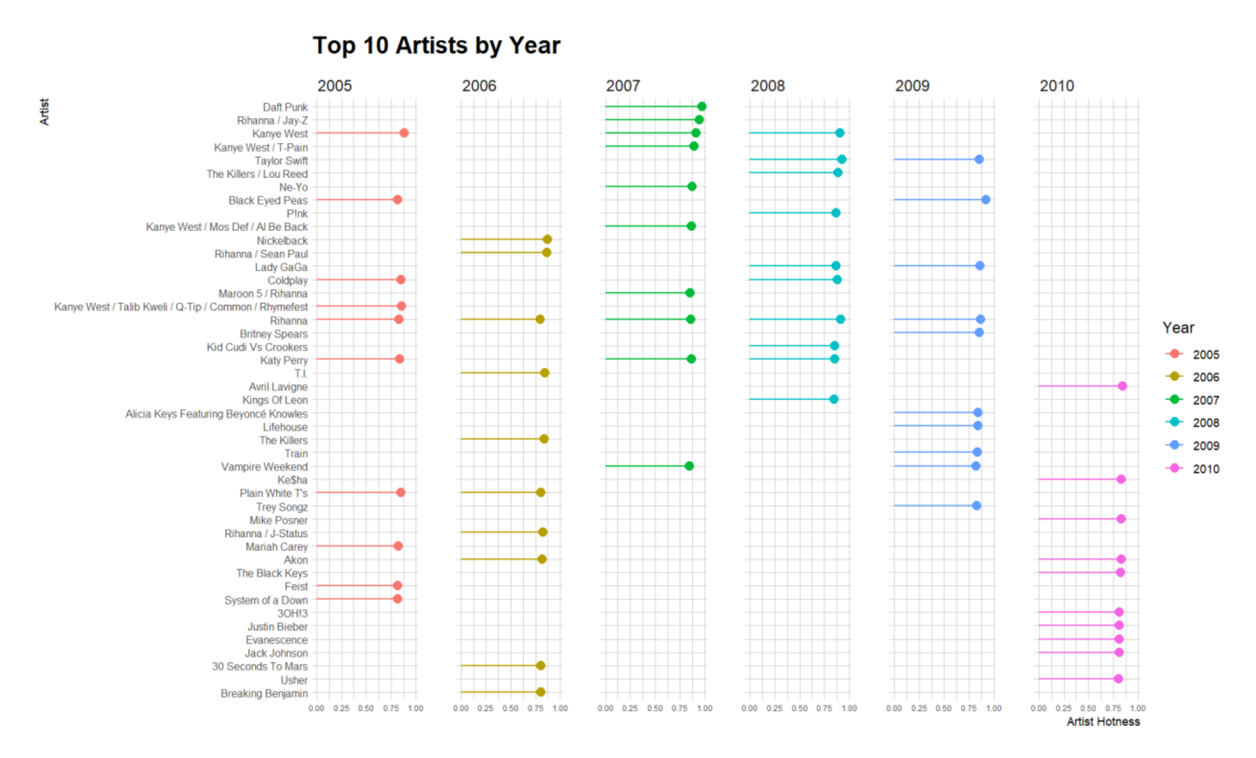


Insight:
In Chart 1, the average song popularity standard has risen, while both average artist familiarity and hotness have gradually declined. The average artist familiarity remains slightly above 0.77, the average song hotness is also 0.77, and the average artist hotness is just below 0.67.

Why this chart?
The line plot effectively illustrates trends and long-term patterns, offering a clear perspective on artist and song performance over multiple years. This provides valuable insights into the evolving standards for selecting artists based on familiarity, hotness, and song popularity.

Call to Action:
Based on Chart 1, artists should ideally have an average familiarity and song hotness above 0.77, and an artist hotness above 0.67. Adhering to these criteria will align the lineup with current trends, increasing audience engagement and turnout. This approach ensures the selection of top-performing artists for the concert.

Chart 2: Lollipop Plot

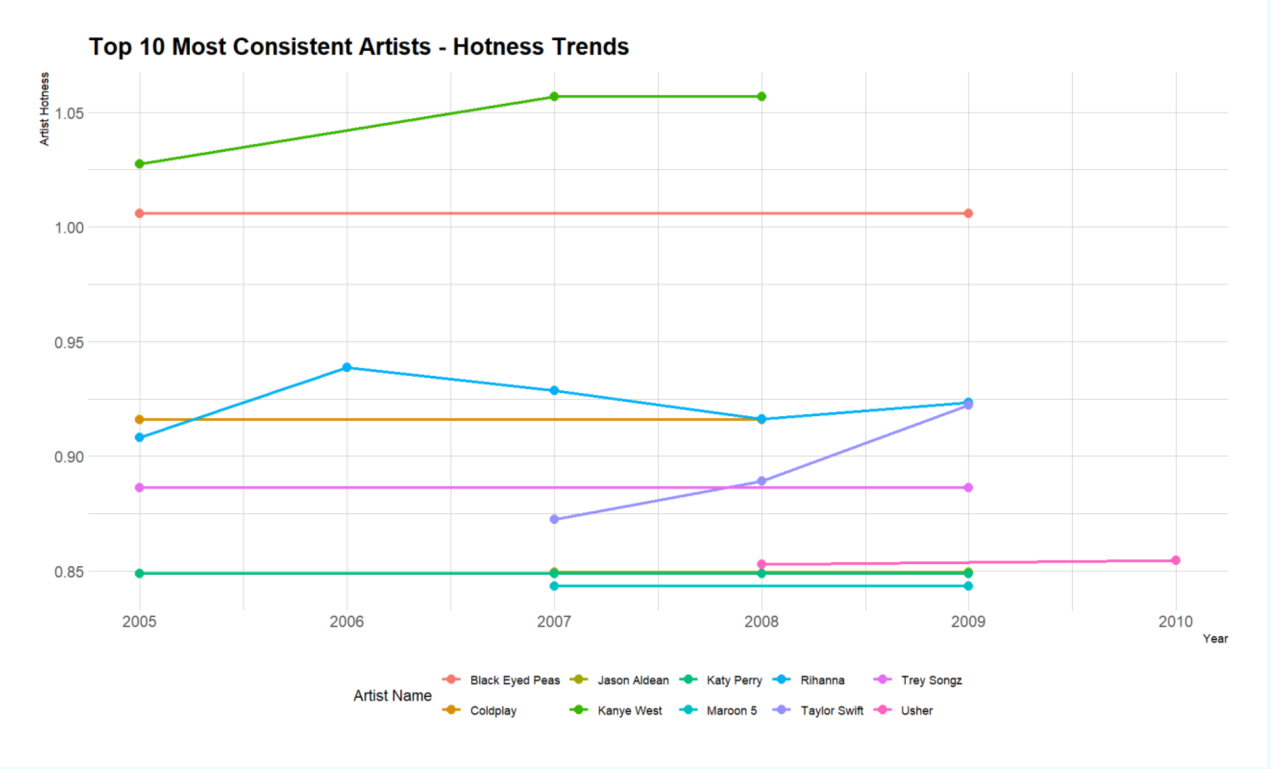


Insight:
Artists such as Rihanna, Katy Perry, and Plain White T's consistently rank in the top 10 for artist hotness, making them strong contenders for selection. This trend helps identify the best-performing candidates.

Why this chart?
A lollipop plot effectively compares multiple rankings side by side, providing a clear visual representation of top-performing artists. This makes it easier to identify the best choices.

Call to Action:
Based on the plot, it is recommended to prioritise artists who have appeared in the top 10 at least twice. This approach helps maximise audience engagement by selecting consistently popular performers.

Chart 3: Line Plot

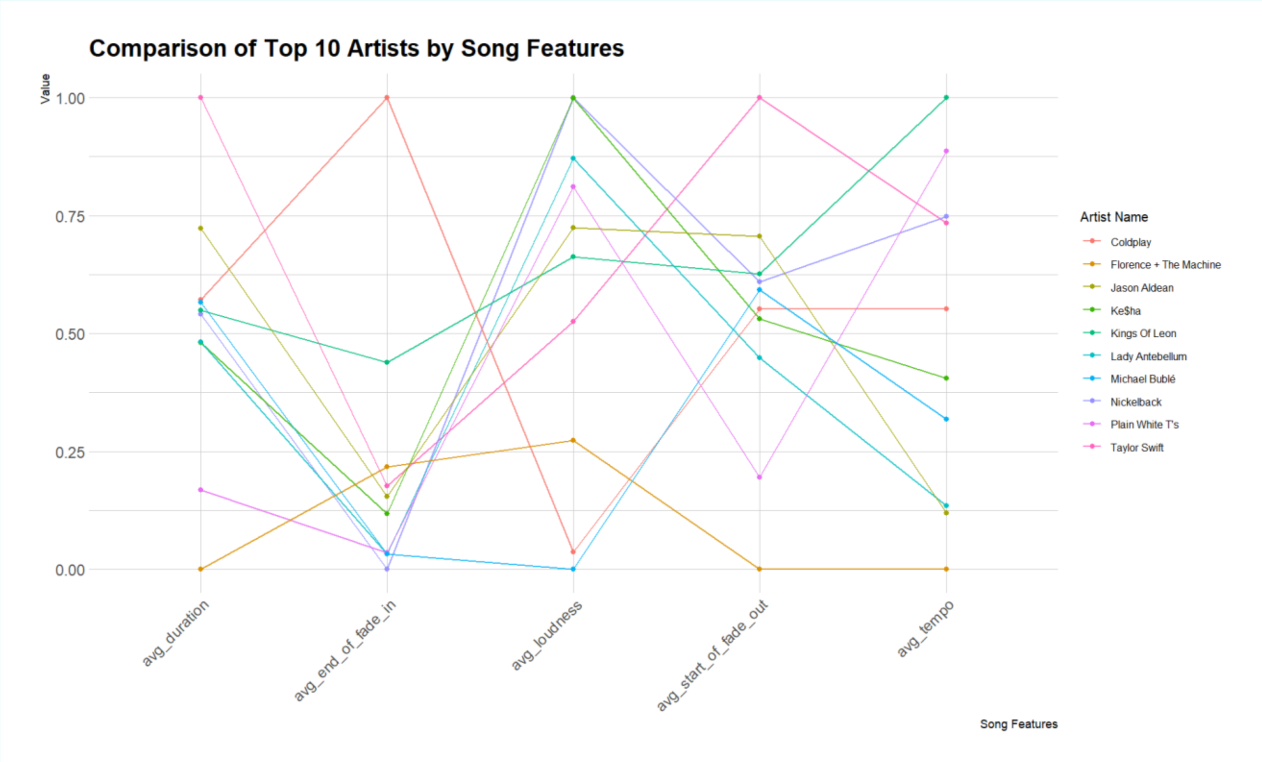


Insight:
Artists such as Black Eyed Peas, Jason Aldean, Trey Songz, Kanye West, and Maroon 5 demonstrate consistent long-term appeal based on their artist familiarity. Katy Perry and Taylor Swift have seen a rise in fame, suggesting they should be prioritised when selecting artists. Meanwhile, Rihanna's notoriety has gradually declined; however, she should still be considered due to her high artist familiarity.

Why this chart?
The line plot effectively illustrates clear trends and long-term patterns, making it easier to identify artists with sustained popularity over time.

Call to Action:
The data suggests prioritising artists with increasing or consistently high audience familiarity. This approach ensures a strong, recognisable concert lineup, minimises the risk of selecting artists with fluctuating popularity, and enhances ticket sales and audience retention.

Chart 4: Parallel Coordinates Plot

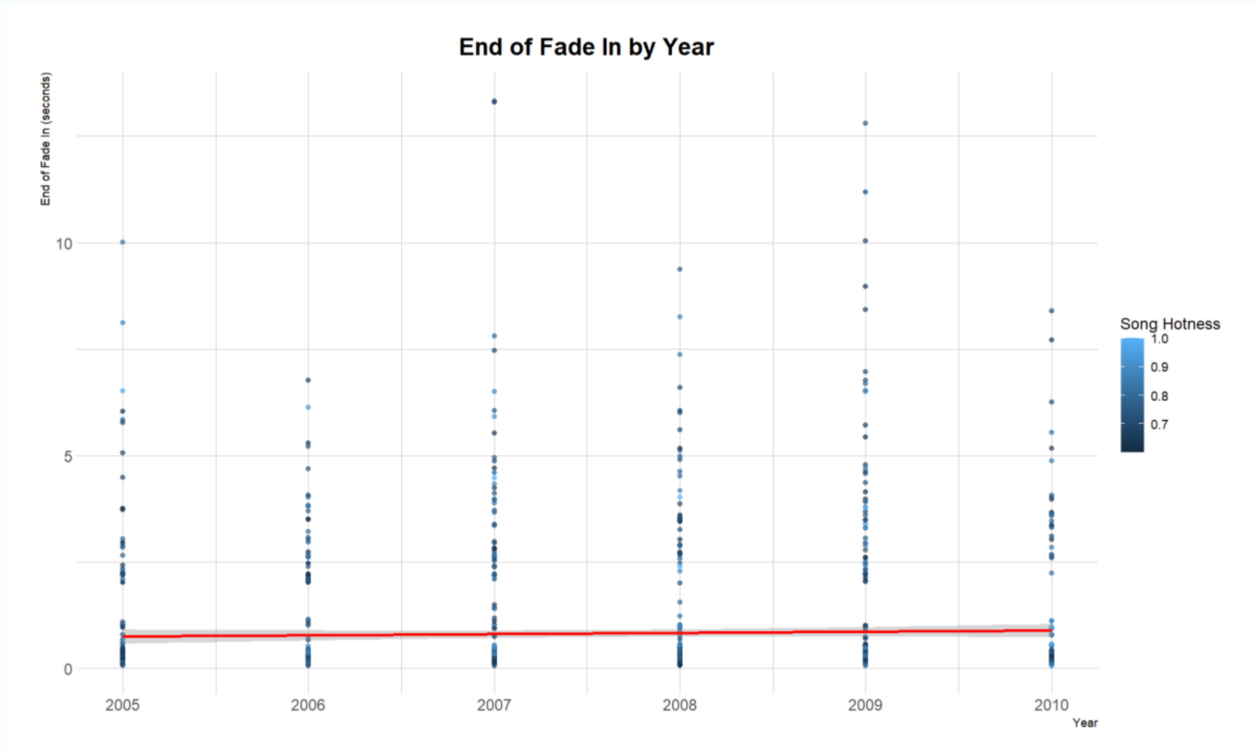


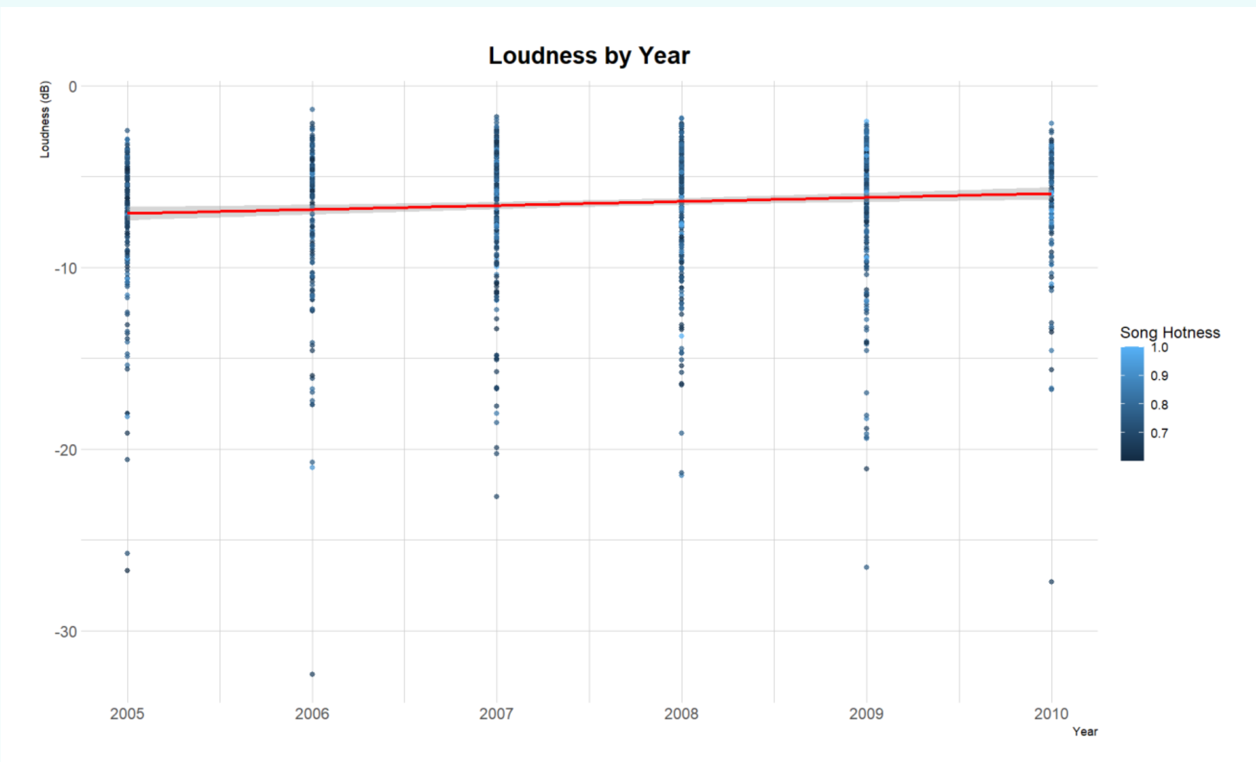
Insight:
Chart 4 compares the top 10 most popular artists based on key song features, including duration, end of fade in, loudness, start of fade out, and tempo. The distribution of these features is well-balanced across the artists, providing a diverse mix of song characteristics.

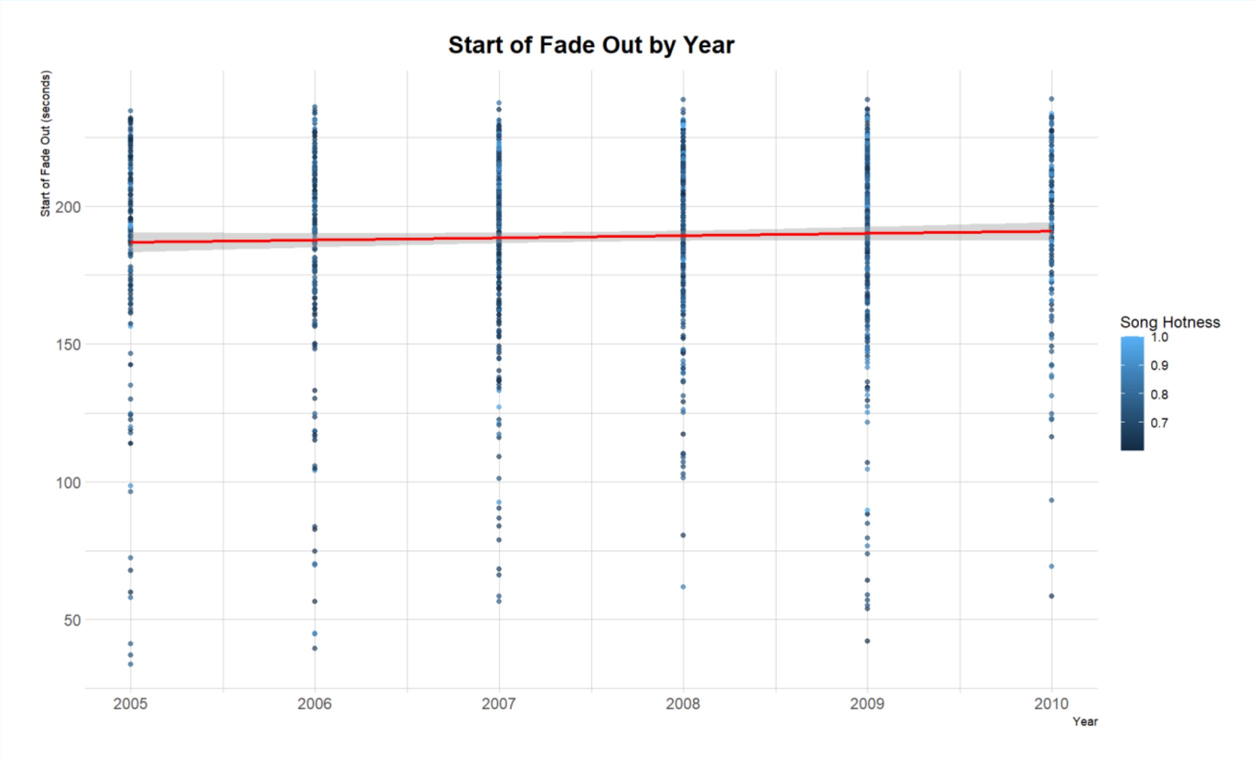
Why this chart?
Parallel coordinate plots effectively display multiple variables simultaneously, making it easy to compare trends across different artists. This visualisation offers a comprehensive overview of each artist's song features, ensuring a well-rounded and dynamic setlist.

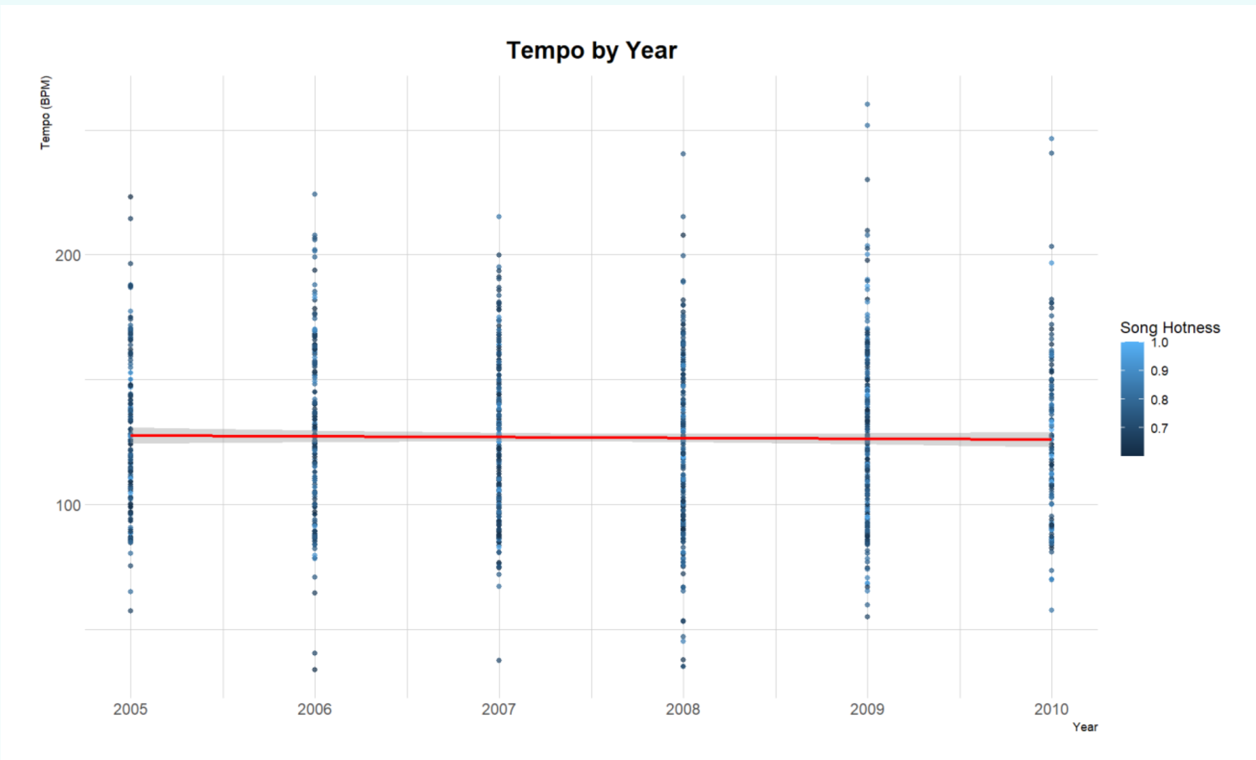
Call to Action:
The chart suggests inviting these artists to create a balanced and engaging concert setlist. This insight supports selecting the most suitable roster for a successful event.

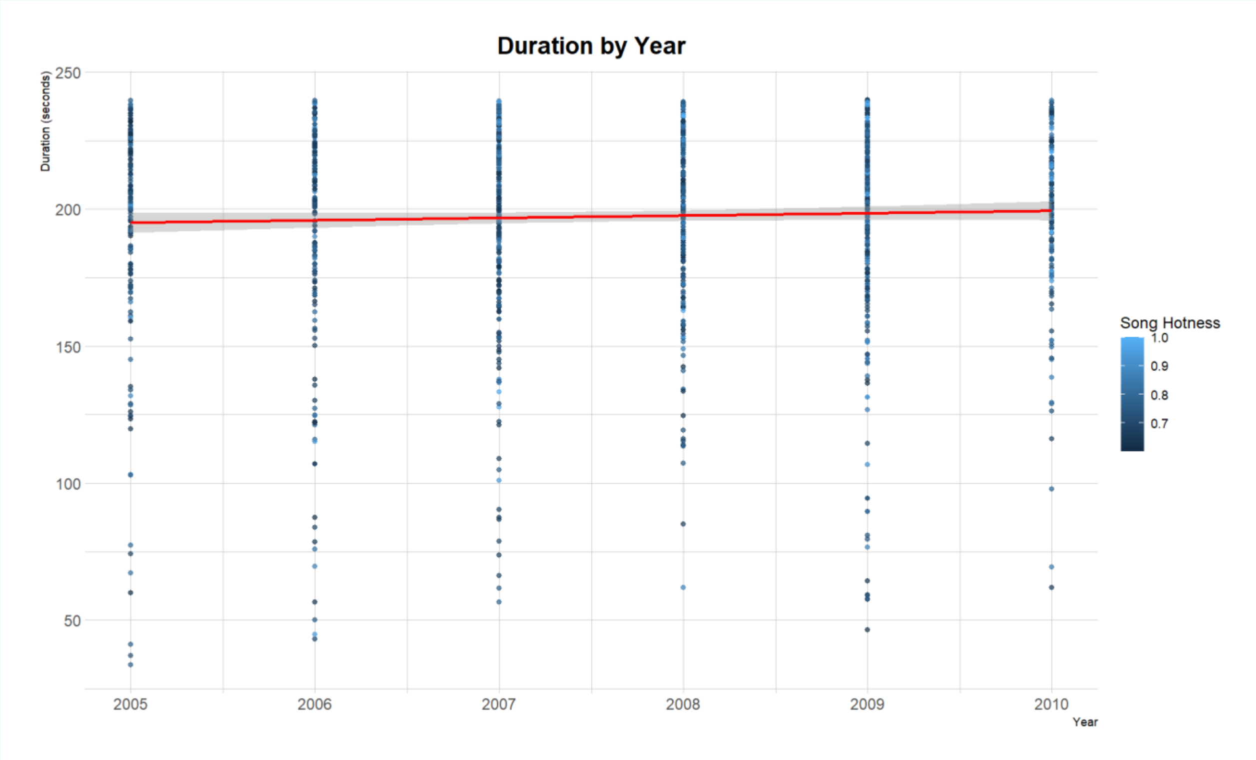
Chart 5-9: Line Plot






Start of Fade Out by Year
Start of Fade Out (seconds)
200
150
100
50
2005
2006
2007
2008
2009
2010
Year
Song Hotness
1.0
0.9
0.8
0.7


Tempo by Year
Tempo (BPM)
200
100
2005
2006
2007
2008
2009
2010
Year
Song Hotness
1.0
0.9
0.8
0.7

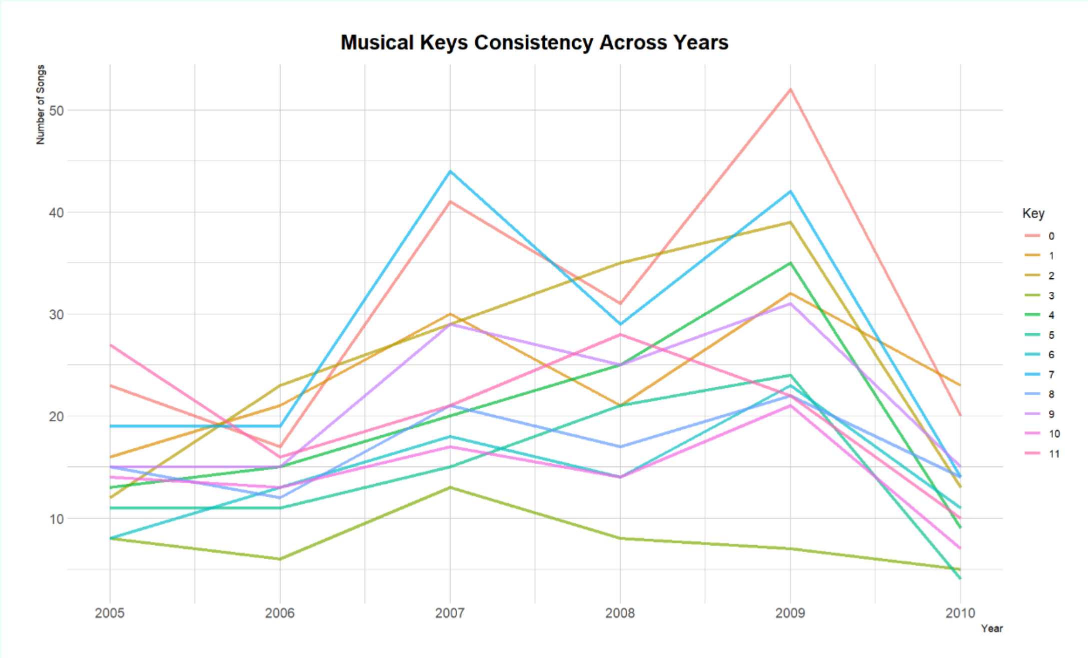


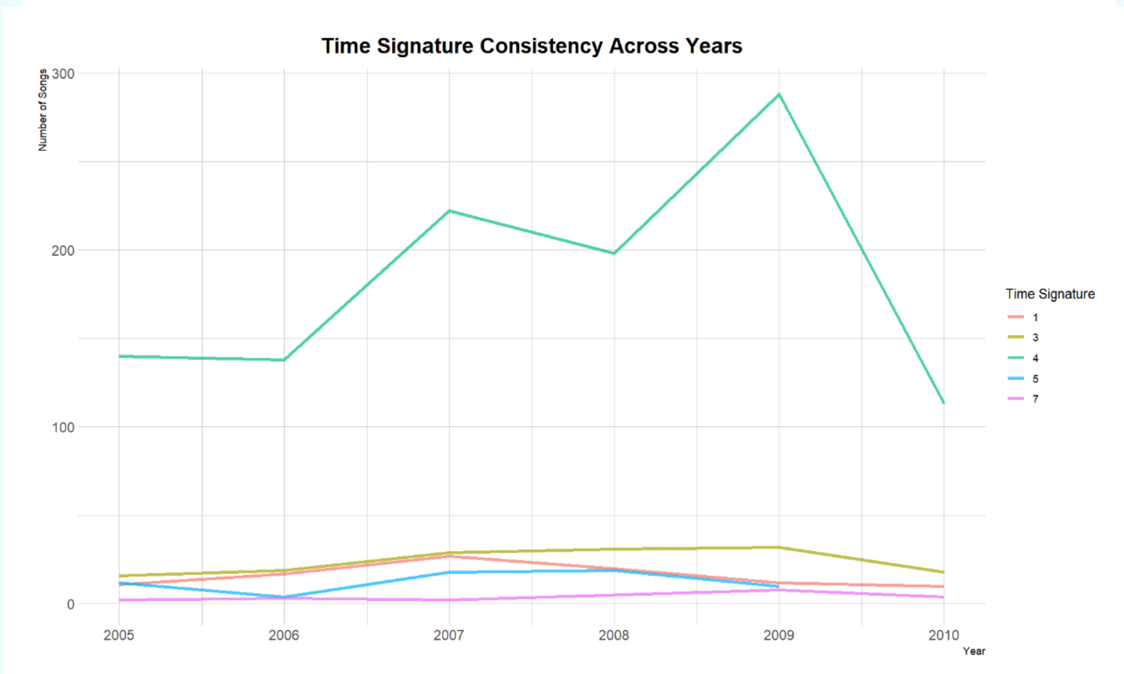
Insight:
Charts 5 to 9 illustrate industry-standard song structures over the past five years. Both end of fade in and tempo have remained consistent over time. The most common format for popular songs includes an end of fade in duration between 0 to 2.5 seconds and a tempo ranging from 75 to 150 BPM. The most used format in popular songs have a loudness between -3 to -10 dB, start of fade out duration 175 to 240 seconds and duration is also between 175 to 240 seconds.

Why these charts?
The line plot effectively highlights trends and long-term patterns in music production, offering a clear understanding of how industry standards have evolved over the years.

Call to Action:
Artists should consider these insights when producing music. By optimising fade-in, fade-out, loudness, tempo, and song duration based on industry trends, they can enhance audience engagement and increase the likelihood of creating a hit song.

Chart 10-11: Parallel Coordinates Plot



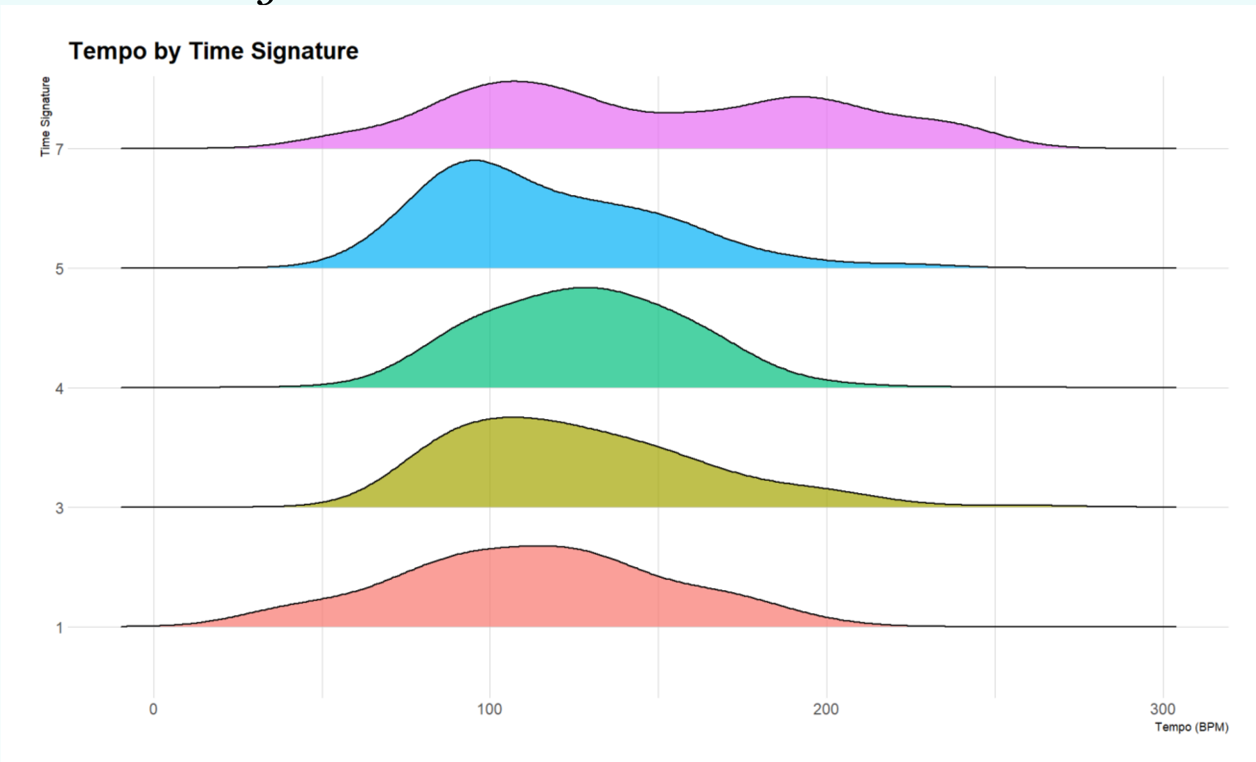


Insight:
Chart 10 shows that all keys, except key 3, exhibit a positive trend peaking in 2009 before experiencing a sharp decline. In contrast, key 3 follows a steady downward trend over the years. Chart 11 further reveals that the most popular songs predominantly use a time signature of 4.

Why this chart?
Parallel coordinates effectively highlight multiple trends, making it easier to visualise how each key and time signature has evolved over time.

Call to Action:
Based on the chart, artists should consider using key 0, as it demonstrates a consistent upward trend without significant drops. Additionally, incorporating time signature 4 is advisable, as it is the most used in popular songs. Adopting these elements can enhance audience engagement by aligning with these current song format trends.

Chart 12: Ridgeline Plot

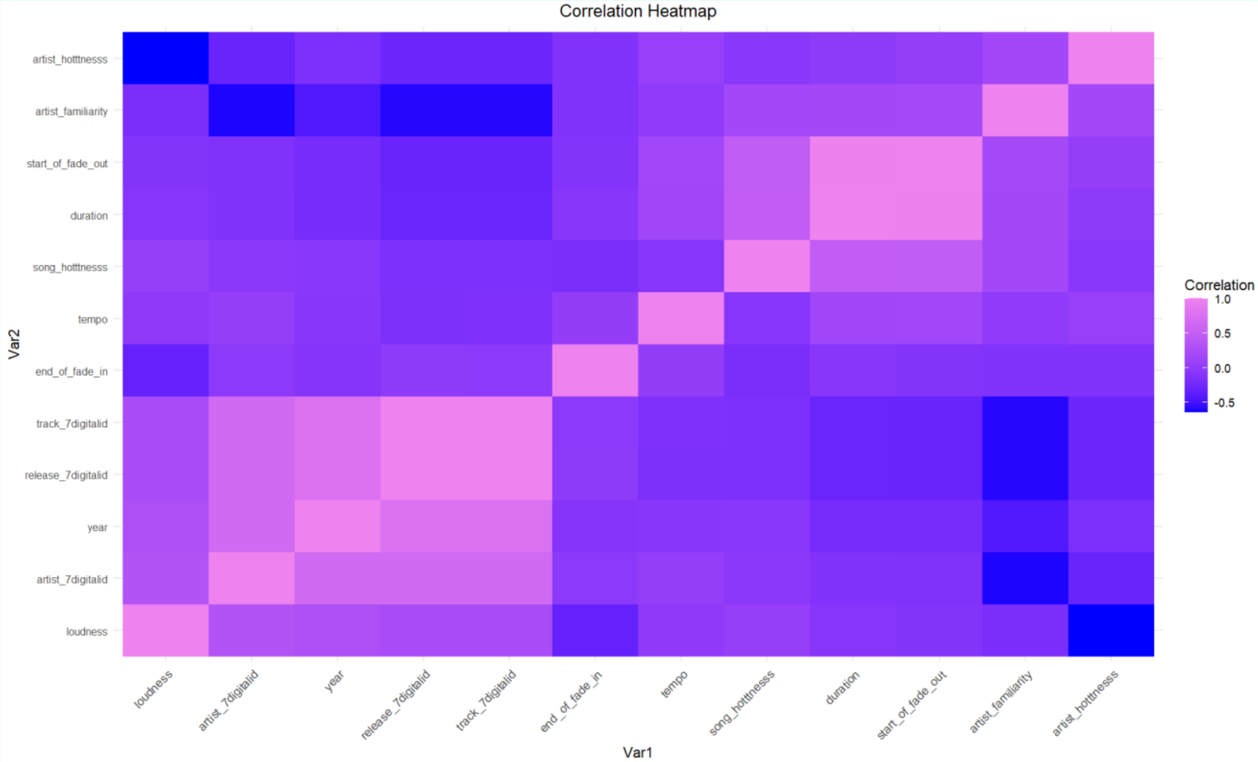


Insight:
From Chart 12, time signatures 5, 3, and 1 are most paired with a tempo of 100 BPM. Time signature 4 is typically associated with 125 BPM, while time signature 7 shows two peaks—one slightly above 100 BPM and another just below 200 BPM.

Why this chart?
Ridgeline plots effectively highlight distributions across multiple categories, making them ideal for visualising the most common tempo and time signature pairings.

Call to Action:
Artists should carefully choose their tempo based on the time signature to optimise their song's quality and enhance audience engagement.

Chart 13: Correlation Matrix

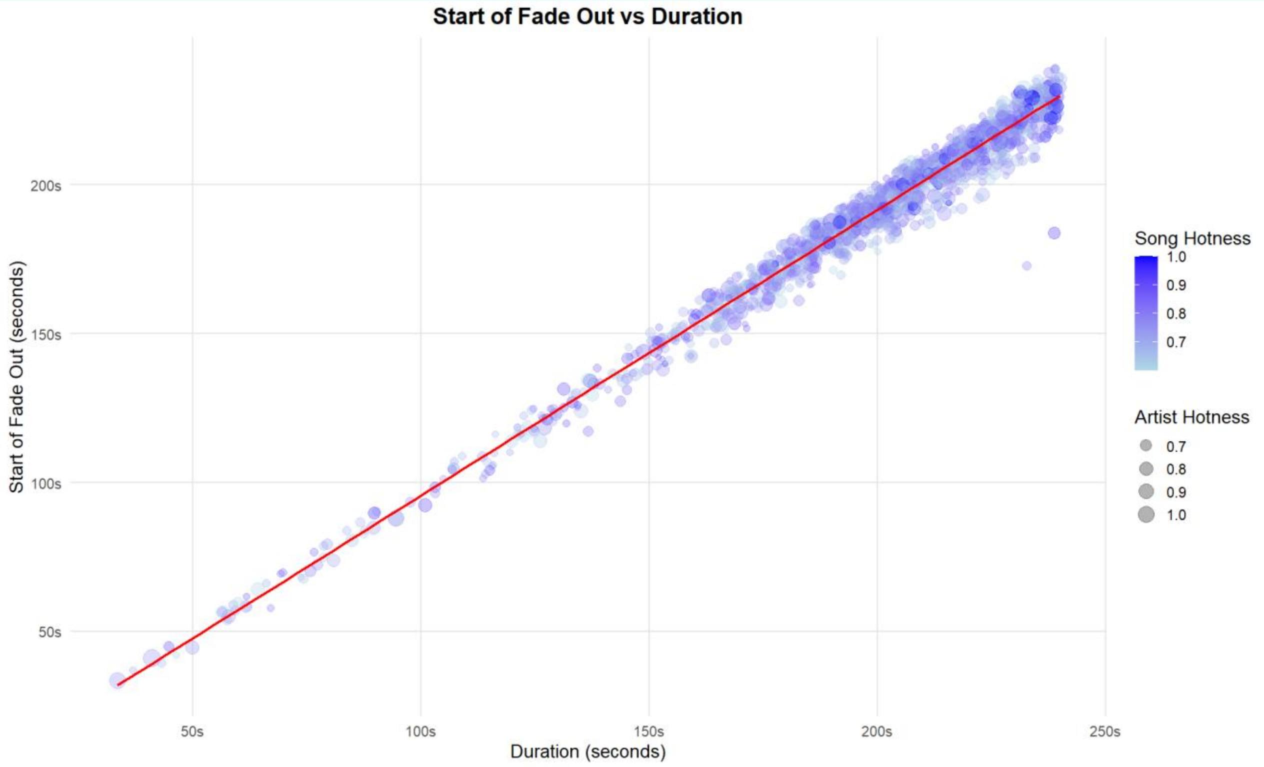


Insight:
Chart 13 shows that most song features have weak correlations with each other. However, two notable correlations emerge: a positive relationship between duration and start of fade out, and a negative relationship between artist hotness and loudness.

Why this chart?
A correlation matrix effectively highlights strong relationships between variables. This visualisation helps analyse patterns within the Million Song Dataset, providing insights into how different features interact.

Call to Action:
Artists should take note of the correlation between duration and start of fade out when crafting their songs. Aligning with this trend may enhance performance quality and improve audience reception.

Chart 14:



Insight:
Chart 14 illustrates a strong correlation between the duration of a song and the timing of its fade-out. As song duration increases, the fade out start time also increases. This suggests that trending artists may follow a specific pattern when creating popular songs.

Why this chart?
Scatterplots are ideal for showcasing relationships between two continuous variables, making it easy to visualise how one factor influences the other. This helps ensure that songs fit within setlist timing while maintaining audience engagement.

Call to Action:
The chart suggests that artists should consider this pattern when crafting their songs. By incorporating this insight, artists can use these optimal metrics to create or identify songs that align with audience preferences, which will enhance concert playlists and improve the overall musical experience for listeners.

Chart 15:

Insight:
Chart 15 showcases the top 10 artists recommended for the concert, along with their 4 most optimal songs to perform on stage.

Why this chart?
The circular bar plot effectively visualises multiple variables in an engaging and clear manner. By minimising visual clutter, it allows for easy comparison of data, making patterns stand out, especially when dealing with many artists and songs.

Call to Action:
Based on this analysis, we recommend inviting these 10 artists and including these songs in their setlists. The chart suggests that these artists and their chosen songs are likely to deliver the most crowd-pleasing concert experience.

CONCLUSION

Through a series of insightful visualisations, including line plots, lollipop charts, parallel coordinates, correlation matrices, and circular bar plots, the analysis recommends the top 10 artists, their hit songs, and optimal song combinations and metrics based on current music trends. The recommendations highlight the most consistently popular artists over time. These suggestions will ensure a high-energy, well-balanced concert lineup that maximises both fan enjoyment and business success for the United Group. By leveraging these data-driven insights, this report empowers the United Group to design concerts that resonate deeply with contemporary audiences.